

Production Server Deployment Guide

Deploying the CEYPETCO application to a live production server involves three main steps: deploying the Database, deploying the Backend (Node.js API), and deploying the Frontend (React SPA).

Here is the step-by-step process:

Step 1: Database Deployment (MySQL)

You need to transfer your local `ceypetco_website` database (including the table, data, and views) to the live server.

- Export the Local Database:**
Open your MySQL client (like DBeaver or MySQL Workbench) on your local machine and use the "Export" or "Dump" tool to generate a `.sql` file containing both the structure and data of the `ceypetco_website` database.
- Create the Live Database:**
On your production server (e.g., via cPanel MySQL Databases), create a new database (e.g., `cpc_website_db`) and a new database user with a strong password. Assign all privileges to that user.
- Import the Data:**
Upload and import the `.sql` file into the new live database using phpMyAdmin or the command line.

Step 2: Backend Deployment (Node.js API)

The backend handles all the data, uploads, and security, and must run constantly in the background.

- Upload Files:**
Upload the contents of the `backend/` folder to your server.
- Install Dependencies:**
Open a terminal on your server (via SSH or cPanel Terminal), navigate to the backend folder, and run:

```
[!WARNING]
Do NOT upload the local node_modules folder or .env file.
```

```
npm install --production
```

- Configure Environment Variables:**
Create a new `.env` file on the production server. Update it with your live settings:

```
PORT=5000
CLIENT_URL=https://ceypetco.gov.lk
JWT_SECRET=<your_strong_random_secret_here>
JWT_EXPIRES_IN=24h
NODE_ENV=production
DB_HOST=127.0.0.1
DB_PORT=3306
DB_USER=<production_db_user>
DB_PASSWORD=<production_db_password>
DB_NAME=<production_db_name>
```

- Start the Server Continuously:**
 - If using a Linux VPS (Ubuntu/Cloud): Use PM2 to keep it running forever.

```
npm install -g pm2
pm2 start src/server.js --name "ceypetco-api"
pm2 save
```

- If using cPanel: Go to "Setup Node.js App", select Node v20+, point the Application root to your backend folder, and click "Start App".

Step 3: Frontend Deployment (React Web App)

The frontend is just a set of static files (HTML, CSS, JS) that need to be built locally and then uploaded.

- Configure API URL:**
In your local `frontend/` folder, create or edit `.env.production` and set the live API URL:

```
VITE_API_BASE_URL=https://api.ceypetco.gov.lk/api
```

- Build the Application:**
Run the following command on your local machine:

```
npm run build
```

This will generate a `dist/` folder containing the highly-optimized production files.

- Upload to Web Server:**
Upload the *contents* of the `dist/` folder (not the folder itself) to the public web root of your domain on the server (usually `public_html` in cPanel, or `/var/www/html/` in Linux).
- Configure SPA Routing (Important!):**
Because this is a Single Page Application (SPA), if a user refreshes the page on `ceypetco.gov.lk/about`, the server must redirect them back to `index.html`.
 - For Apache (cPanel): Add this `.htaccess` file in your `public_html` folder:

```
<IfModule mod_rewrite.c>
  RewriteEngine On
  RewriteBase /
  RewriteRule ^index\.html$ - [L]
  RewriteCond %{REQUEST_FILENAME} !-f
  RewriteCond %{REQUEST_FILENAME} !-d
  RewriteRule . /index.html [L]
</IfModule>
```

- **For Nginx (VPS):** Ensure your server block includes:

```
location / {
    try_files $uri $uri/ /index.html;
}
```

Step 4: Final Checks

- Ensure an SSL Certificate (HTTPS) is active for both `ceypetco.gov.lk` and `api.ceypetco.gov.lk`.
- Ensure the `backend/uploads/` directory on the live server has correct write permissions (e.g., `chmod 755`) so admins can upload files and images.